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Monitoring mangrove forest changes and estimating carbon stock in Bac Lieu, Vietnam from 2014 to 2023 by using remote sensing

Hong Minh Nguyen
College of Environment and Natural Resources, Can Tho University
Can Tho, Vietnam
hongminh122003@gmail.com

Thi Phuong Linh Vo
College of Environment and Natural Resources, Can Tho University
Can Tho, Vietnam
vtplinh@ctu.edu.vn

Quoc Thanh Vo[1]
College of Environment and Natural Resources, Can Tho University
Can Tho, Vietnam
quocthanh@ctu.edu.vn
ORCID 0000-0001-5940-1798

Mangrove forests are a critical wetland ecosystem in minimizing CO_2 concentrations in the atmosphere and promoting sustainability. Bac Lieu province, Vietnam, possesses extensive mangrove forest areas. However, mangroves have recently suffered from human activities and climate change, leading to forest degradation and loss. The conventional forest monitoring method is ineffective, especially in large-scale assessments. This study utilized remote sensing to analyze mangrove forest changes and estimate carbon stock in Bac Lieu province from 2014 to 2023 to propose effective management strategies and develop carbon offsets and carbon credits. Results indicate that the Maximum Likelihood Classification (MLC) algorithm outperforms the Random Forest (RF) algorithm in classification accuracy. Mangrove forest expanded from 5,101 ha in 2014 to 5,581 ha in 2023, with expansion primarily in the Dong Hai district. The estimated carbon stock fluctuates in a range of 58 to 280 KT, and no significant deviation compared to the field-based estimation method. Land-use conversion to aquaculture and policy constraints are the main causes of mangrove degradation in Bac Lieu Province. To address the challenges facing mangrove ecosystems in Bac Lieu Province, a multi-faceted approach integrating socio-economic, sustainability, and technological strategies is essential.

Keywords: carbon credits, land-use conversion, forest restoration.

I. Introduction

Mangroves are a crucial type of wetland ecosystem predominantly found in coastal regions. Although wetlands cover only 5 to 8% of the Earth's surface, they are estimated to store approximately 35% of the global carbon stock, highlighting their role in climate change mitigation [1][2][3]. This carbon storage capacity offers potential for carbon credits, incentivizing organizations and businesses to reduce emissions and promote sustainable development [3][4].

Despite their ecological value, mangroves face increasing threats from human activities and climate change. Land-use conversion to aquaculture has contributed to deforestation and environmental pollution [5]. For instance, the Philippines has lost approximately 50% of its mangrove forests over the past 30 years, and China has experienced a 40% decline in its mangroves [6][7].

Monitoring changes in forest areas and biomass using traditional methods, such as field surveys, is often time-consuming, labor-intensive, and costly [2][8][9]. In contrast, advancements in remote sensing and Geographic Information Systems (GIS) provide efficient solutions for rapid and accurate periodic observations of the Earth's surface and precise analysis of forest areas. Numerous classification algorithms have been developed to analyze land cover classes. Commonly used algorithms, such as Maximum Likelihood Classification (MLC), are standard approaches due to their straightforward implementation and high accuracy [5],[10] [11]. Additionally, Random Forest (RF), Support Vector Machine (SVM), and Decision Tree are widely employed for classification, with RF often evaluated as more effective [9][4][12].

Numerous studies have demonstrated the effectiveness of remote sensing and GIS techniques in monitoring forest cover changes. For instance, in Mozambique's Zambezi Delta, which boasts Africa's largest mangrove forest area, remote sensing has been effectively utilized to monitor both mangrove extents and carbon stocks with high accuracy [13]. Similarly, in Malaysia's Iskandar region, a rapidly growing economic zone in southern Johor, MLC was applied to monitor mangrove areas, achieving over 80% accuracy and surpassing the performance of the SVM algorithm [14].

Furthermore, accurately measuring forest biomass through in situ methods involves destructive sampling, such as cutting down trees, which is labor-intensive, costly, and restricted to small areas. In contrast, remote sensing offers rapid, accurate, and effective monitoring over extensive regions. The Normalized Difference Vegetation Index (NDVI) is widely employed for estimating forest biomass [2][15][16]. This index has been successfully used to estimate mangrove biomass and carbon stocks in Ca Mau Province, Vietnam, with high accuracy, demonstrating its potential for broader application [8].

In Vietnam, remote sensing and GIS are effectively utilized for monitoring mangrove forest areas. However, their application for comprehensive forest inventories, particularly for estimating biomass and carbon sequestration, remains limited. Bac Lieu Province, situated in the Mekong Delta, encompasses over 4,000 hectares of mangroves rich in biodiversity [17]. Nevertheless, industrialization and modernization have hastened deforestation and intensified climate change impacts, resulting in forest degradation and loss [3][17][18][19]. Although Bac Lieu conducts annual forest inventories using conventional methods, the use of GIS and remote sensing for assessing forest areas and biomass is

[1] Corresponding author: QT. Vo

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constrained by a lack of specialized personnel and optimized methodologies tailored to the region.

This study employs remote sensing and GIS techniques to construct a forest change map for Bac Lieu Province over the period 2014–2023 and to estimate carbon sequestration. The research aims to propose effective management strategies and assess their potential for generating carbon credits, thereby contributing to "Net-Zero" objectives.

II. Materials and Methods

A. Study area

Mangrove forests are mainly distributed in coastal areas [20] [21] [22]. Therefore, this study area is located in three Bac Lieu coastal districts, including Dong Hai, Hoa Binh, and Bac Lieu City (Fig. 1).

Fig. 1. Administrative map of Bac Lieu Province

B. Materials

1) Ground truthing data

The study carried out a ground survey in the study area to collect samples for classification and accuracy assessment. A total of 285 samples were collected, representing 4 land cover classes: mangrove forest, water bodies, built-up areas, and other vegetation. 60% of these points were used as training input data for classification models, while the remaining 40% were used for accuracy validation.

2) Remote sensing imagery

Google Earth Engine (GEE) is a cloud-based geospatial analysis platform that enables users to visualize and analyze satellite images of the Earth surface. In this study, all Landsat 8 images with a 30 x 30 m spatial resolution in 2014, 2019, and 2023 were collected from the USGS Landsat 8 Level 2, Collection 2, Tier 1. The preprocessing includes cloud and cloud shadow removal, and atmospheric correction was implemented using JavaScript in GEE. Finally, a median composite approach was applied to ensure representative images of land cover conditions for each selected year.

C. Methods

1) Classification algorithms

a. Random Forest algorithm (RF)

RF was applied to determine classification thresholds based on the NDVI and Combined Mangrove Recognition Index (CMRI) as input data. The optimal number of decision trees of RF in this study was determined automatically through a Python script in QGIS 3.34.11.

- NDVI equation:

$$NDVI = \frac{B_{NIR} - B_R}{B_{NIR} + B_R} \quad (1)$$

Where B_{NIR} is the infrared band; B_R is the red band.

- CMRI equation:

$$CMRI = NDVI - NDWI \quad (2)$$

- NDWI is the normalized difference water index. Equation:

$$NDWI = \frac{B_G - B_{NIR}}{B_G + B_{NIR}} \quad (3)$$

Where B_{NIR} is the infrared band; B_G is the green band.

b. Maximum Likelihood Classification (MLC) algorithm

The MLC is a supervised classification method. In this study, MLC was applied and compared with RF to determine a more accurate algorithm for land cover mapping.

2) Accuracy assessment

An error matrix, also known as a confusion matrix, represents the discrepancies between classified pixels and ground truth reference data. This method is an effective approach for assessing classification accuracy, incorporating key metrics such as Overall Accuracy (OA), Producer's Accuracy (PA), and User's Accuracy (UA). Additionally, the Kappa coefficient (K) was used to evaluate the accuracy of the classification between the classified results and the reference data [23].

$$OA = \frac{Total\ number\ of\ correctly\ classified\ pixels\ (diagonal)\ x\ 100}{Total\ number\ of\ reference\ pixels} \quad (4)$$

$$UA = \frac{Number\ of\ correctly\ classified\ pixels\ in\ each\ category\ x\ 100}{Total\ number\ of\ classified\ pixels\ in\ that\ category\ (row\ total)} \quad (5)$$

$$PA = \frac{Number\ of\ correctly\ classified\ pixels\ in\ each\ category\ x\ 100}{Total\ number\ of\ classified\ pixels\ in\ that\ category\ (column\ total)} \quad (6)$$

$$K = \frac{(Total\ sample\ x\ Total\ corrected\ sample) - \sum(Column\ total\ x\ Row\ total)}{Total\ sample x^2 - \sum(Column\ total\ x\ Row\ total)} \quad (7)$$

TABLE I. KAPPA COEFFICIENT VALUES FOR ACCURACY ASSESSMENT IN IMAGE CLASSIFICATION [24]

Kappa Range	Strength of agreement
< 0	Poor
0 – 0.2	Slight
0.21 – 0.4	Fair
0.41 – 0.6	Moderate
0.61 – 0.8	Substantial
0.81 – 1	Almost Perfect

D. Carbon stock estimation

Forest carbon stock (C) is estimated based on forest biomass [25].

$$C = TGB\ x\ 0.4 \quad (8)$$

Where C is carbon stock; 0,47 is the average accumulated carbon value of mangrove forests. TBG (Total Biomass Growth) refers to the biomass above ground (Above Ground Biomass - AGB) and below ground (Below Ground Biomass - BGB).

$$TGB\ (ton) = AGB + BGB \quad (9)$$

- Estimation using remote sensing data: This study utilizes a regression model established by Ha (2017) to estimate

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forest biomass by NDVI value. The equation for total biomass:

$$TGB = exp\left(\frac{1}{0.13489+0.00823004*NDVI^{-4}}\right) \quad (10)$$

- Estimating forest biomass from wood volume

- AGB represents all living biomass above the ground, including tree trunks, branches, foliage, and stems

$$AGB = BCEF \times Total\ wood\ volume \quad (11)$$

Where total wood volume is statistically analyzed from field measurement data provided by the Bac Lieu Forest Protection Sub-Department. BCEF (Biomass Conversion and Expansion Factor) is used to convert wood volume to aboveground biomass [26][27].

- BGB consists of all root biomass, excluding fine roots with a diameter of less than 2 mm:

$$BGB = AGB \times 0.2 \quad (12)$$

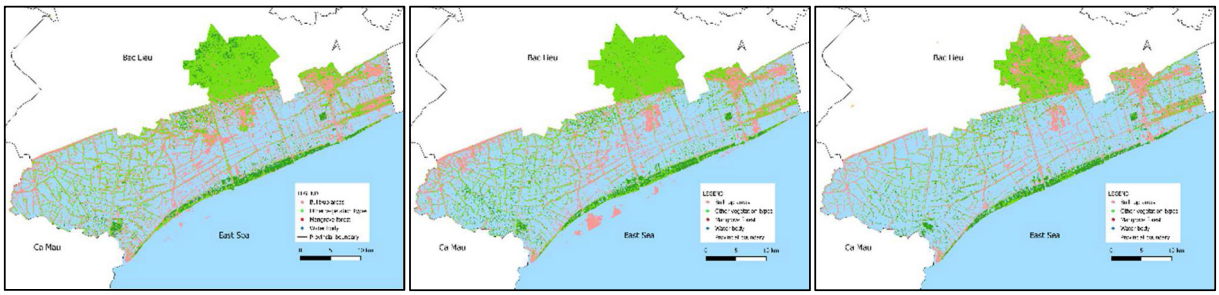
III. RESULTS AND DISCUSSION

A. Land cover mapping

Using RF to determine threshold values for classification based on NDVI and CMRI, the classification accuracy was over 60% with K = 0.46 for NDVI and 53% with K = 0.35 for CMRI. These methods showed limited effectiveness in land cover classification due to spectral confusion with other vegetation classes. In contrast, the Maximum Likelihood Classification (MLC) algorithm achieved higher accuracies: 90% (K = 0.82) in 2023, 88% (K = 0.8) in 2019, and 79% (K = 0.62) in 2014. Consequently, MLC was selected for land cover mapping in this study.

B. Mangrove forest changes

Mangrove forests in Bac Lieu Province are primarily distributed along the coastline and have expanded from 2014 to 2023 (Fig. 2). In 2014, the total mangrove coverage was approximately 5,101 ha. The mangrove area increased to 5,241.78 ha in 2019 and further expanded to 5,581 ha by 2023. This growth reflects the positive impacts of mangrove restoration and conservation efforts along the coastal belt in Bac Lieu Province.



2014 **2019** **2023**

Fig 2. Mapping classification in the study area in 2014, 2019, and 2023.

The total mangrove forest area in Bac Lieu Province increased by 479.61 ha during the period from 2014 to 2023 (Table II). The majority of the expansion occurred in the Dong Hai district, which recorded an increase of approximately 999.36 ha. In contrast, Hoa Binh District experienced a significant decline in mangrove cover, with a total loss of 779.58 ha (Fig 3).

TABLE II. MANGROVE FOREST AREA IN COASTAL DISTRICTS OF BAC LIEU PROVINCE IN THE PERIOD FROM 2014 TO 2023

Unit: ha

Location	2014	2019	2023	Change (2014 – 2023)
Bac Lieu City	516.87	509.67	776.7	+259.83
Dong Hai	1,691.91	2,707.83	2,691.27	+999.36
Hoa Binh	2,892.69	2,024.28	2,113.11	–779.58
Total	**5,101.47**	**5,241.78**	**5,581.08**	**+479.61**

Meanwhile, remote sensing analysis of mangrove changes in Bac Lieu Province from 1988 to 2018 showed the expansion of 1,069 ha, including an increase of 401.5 ha in Hoa Binh District [11]. The primary cause of the reduction in Hoa Binh district during the study period is the high rate of forest land conversion to aquaculture. Therefore, land-use conversion is severely impacting the mangrove forest area in Bac Lieu province [28][29]. Additionally, sediment deficiency in the Mekong Delta has contributed to forest degradation, leading to erosion and landslides along the shoreline [30]. These negative impacts place significant pressure on the mangrove ecosystem.

Fig 1. Changes in mangrove forest area in Bac Lieu Province from 2014 to 2023.

C. Carbon sequestration potential

The estimation of total forest biomass from NDVI values indicates that the mangrove forest in Bac Lieu province ranges from approximately 124,016 tons to 595,833 tons of biomass. Accordingly, the estimated carbon stock is calculated to range from 58,287 tons to 280,041 tons. These

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results show no significant deviation when compared to the wood volume estimation method.

TABLE III. The carbon stock of mangrove forests in Bac Lieu is estimated based on wood volume and remote sensing data

Unit: Ton (t)

Method	Index	Min	Average	Max
Wood Volume	TGB	198,997	386,693	601,413
	C	93,528	181,745	282,664
Remote sensing	TGB	124,016	206,710	595,833
	C	58,287	97,153	280,041
Error (%)		3.9	67.9	0.9

Bac Lieu province has forest biomass averages ranging from 22.3 to 107.2 t/ha. In comparison, Mui Ca Mau National Park, covering 41,862 ha in Ca Mau Province (which borders Bac Lieu province), has estimated AGB between 154.93 ± 61.76 t/ha and 264.86 ± 86.36 t/ha, and BGB from 84.33 ± 35.15 t/ha to 130.56 ± 46.39 t/ha [31]. Despite having a smaller forest area, Mui Ca Mau National Park possesses higher biomass, reflecting greater carbon storage potential compared to Bac Lieu Province. This difference is primarily attributed to tree density and age [31][32][8]. The strict conservation and restoration efforts in this national park, recognized as a Ramsar site, have contributed to higher tree density and age. Similarly, in U Minh Thuong National Park (Kien Giang province), the total AGB of young acacia forests under 10 years old is 65.63 t/ha, and for trees older than 10 years, it is 89.98 t/ha [33]. In conclusion, effective forest management and conservation can increase tree density and limit deforestation, thereby enhancing forest biomass and carbon storage.

D. Proposed management strategies

The research indicates that the coastline of Bac Lieu Province is experiencing significant mangrove loss along the shores of all three coastal districts. To address this issue, monitoring and protecting shorelines are essential to detect and promptly respond to negative impacts on the coastline and mangrove forests [34]. GIS techniques and remote sensing enable the detection of shoreline and forest changes in near real-time [34][35]. Additionally, implementing construction solutions such as dikes and revetments has shown positive results in protecting the coastline [35].

The management and conservation of the mangrove system require strong collaboration among the government, local communities, and other stakeholders [36]. Indonesia, which has the largest area of mangrove forests in the world, employs co-management strategies, establishing networks and systems that connect the government, donors, local communities, and other stakeholders [9][37]. Similarly, Thailand has community-based forest management intertwined with cultural practices and traditional knowledge [38]. Developing and implementing clear and transparent policies that define the rights and responsibilities of all stakeholders in mangrove forest management is crucial. Such policies not only help minimize conflicts of interest but also enhance accountability and cooperation among parties in protecting forest resources.

Bac Lieu Province is a major center for shrimp aquaculture in Vietnam, leading to widespread conversion of forest land into aquaculture [11]. Therefore, balancing economic development with ecosystem protection has become a critical issue. The integrated shrimp–mangrove farming system is a viable solution for securing local livelihoods while encouraging community involvement in forest protection [39]. However, promoting eco-certificates such as Naturland or Aquaculture Stewardship Council is necessary to enhance the value of sustainable production and ensure stable income sources for local communities [40][37]. Additionally, the carbon credit market offers further income opportunities, incentivizing farmers to engage in forest conservation [41]. Nevertheless, ensuring a balance between environmental sustainability and economic interests requires a well-defined policy framework.

In summary, protecting the coastline, improving policies, and encouraging community involvement will help ensure the sustainable protection of the mangrove system while supporting local livelihoods.

IV. CONCLUSION

This study applied MLC for mapping due to its high accuracy, exceeding 80%, which outperformed the thresholding of NDVI and CMRI by RF. Mangrove forest areas in Bac Lieu province increased from 5,101 ha in 2014 to 5,581 ha in 2023, primarily expanding in Dong Hai district. The carbon stock, estimated based on NDVI, ranged from 58,287 to 280,041 tons, with differences from the wood volume estimation method ranging from 0.9% to 67.9%, noting a larger discrepancy for the average. These results demonstrate the effectiveness of remote sensing data and GIS techniques for carbon storage estimation. Although mangrove forest areas in Bac Lieu province are increasing, this ecosystem remains affected by natural and human impacts. To ensure sustainable conservation and restoration, it is essential to (1) monitor coastline and mangrove forest changes, (2) develop policies that balance the interests of stakeholders, and (3) promote local livelihoods linked to forest conservation.

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